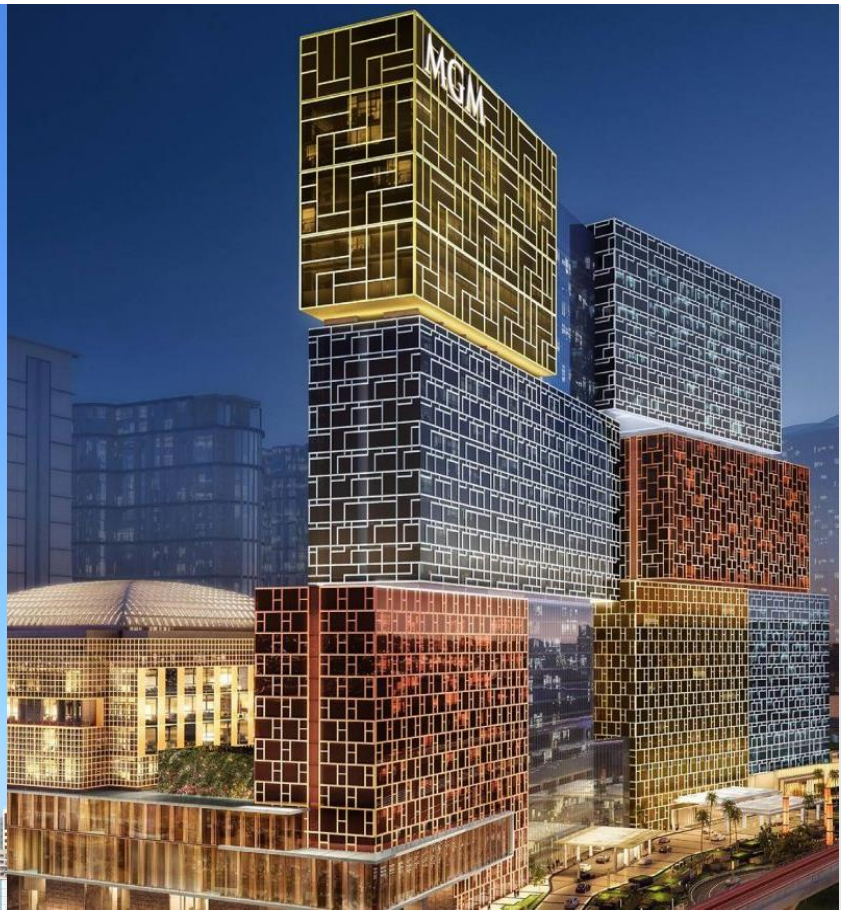
MGM MACAU & COTAI
RESOURCE EFFICIENCY MANAGEMENT REPORT
JULY 2023

Prepared For :

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EXECUTIVE SUMMARY

NV5 has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. NV5 appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

Key Performance Indicators

NV5 reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as NV5's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	2022	Normalized Savings Achieved Up to July 2023	2024	2025
MACAU	2019	Target	-4%	-8%	-10%	N/A	**-13%**
		Actual	**-5.9%**	**-10.8%**	**-5.4%**	-	**-**
COTAI	2019	Target	-4%	-8%	-10%	N/A	**-13%**
		Actual	**-6.4%**	**-16.9%**	**-4.9%**	-	**-**

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2023.

Table 2. Internal Actual Energy Performance Targets for 2025

MGM Properties	Baseline	Actual Saving Target[1]	Savings Achieved Up to July 2023
MACAU	2019	13%	**-4.6%**
COTAI	2019	13%	**-5.1%**

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams.

Highlighted Changes in Operation

MACAU

1. New water-to-water HX was installed for podium MVAC space heating, testing works were conducted since Feb 2023.
2. Grande Praça 7 Units AHU were installed with PICCV.
3. MGM Macau implemented the Hotel Tower Staircase lighting delamping by removing one (1) piece of T5 tube from the existing fluorescent lighting fixture.
4. MGM MACAU replaced 6 units of UPS with high efficiency model.
5. MGM MACAU will replace the rooftop façade LED by 2023.
6. MGM MACAU will upgrade the Lion Backdrop lighting by 2023.

COTAI

1. Grill 58, MDM, Ballroom, Hao Guo kitchen exhaust fan steps speed control program and graphics were updated in BMS.
2. MGM COTAI hotel towers façade lighting control has transferred to IT team, it will be dimmed down from 02:00 – 17:30 instead of existing switch OFF. This is to reduce the LED failure rate and extend the LED life time.
3. MGM COTAI Mansion AHU-6FM-01, 03, 08 has utilized the LoRa sensor data to reset the RAT setpoint.
4. MGM COTAI installed two (2) additional sets of E-bus charging stations in car park.
5. MGM COTAI Kitchen Exhaust Fan operation hours have been fine-tuned according to the latest F&B team operation needs.

Consumption Intensity

The 2023 YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m²)	% Change Compared to 2022
MACAU	**26.3**	**+7.9%**
COTAI	**26.7**	**+12.2%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m³/ m²)	% Change Compared to 2022
MACAU	**0.25**	**+29.2%**
COTAI	**0.19**	**+55.6%**

ISO 50001 Certification Audit

MACAU

The HKQAA 2023 surveillance audit was completed on 28 April 2023. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

COTAI

The HKQAA 2022 recertification audit was completed on 30 September 2022. There is no nonconformity (NC) and areas for improvement (AFI) from auditors. 2023 surveillance audit will be scheduled on 12 October 2023.

Target Milestones in Year 2023

The following activities are to be completed in year 2023:

Table 5. MGM MACAU Year 2023 Projects

Item	Initiatives	Status
Electricity		
1	Secondary Chilled Water Pump Retrofit (Phase 1)	Postpone to 2024
2	Cooling Tower Fan Retrofit (Phase 1)	On-going
3	PICCV Implementation 2023 (Grande Praça 7 Units)	Completed
4	Neon Light LED Replacement (Phase 3)	Postpone to 2024
5	Hotel Tower Staircase Lighting Delamping	Completed
6	Rooftop Façade LED Replacement	On-going
7	Lion Backdrop Lighting Upgrade	On-going
LPG		
8	Natural Gas Changeover Project	On-going

Table 6. MGM COTAI Year 2023 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Plant Controls Optimization (CHWP DP Setpoint Reset, CHWST Setpoint Reset)	On-going
2	PICCV Implementation 2023 (12 Units)	Postpone to 2024
3	KEF Steps Speed Control Optimization (Vista, Ball Room, MDM, Hao Guo, Five Foot Road, Chún, Coast)	On-going
4	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)	On-going
5	Centralized RO Pump Set Replacement (10 units)	Completed
6	Booster Pump Set Replacement (2 units)	Completed
Water		
7	Extension of B2 Carpark CDP Water Recycling (Phase 5 to 7)	On-going
8	Spectacle Façade Rainwater Harvesting for CT Make Up	TBC

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ACKNOWLEDGMENT

NV5 has been working with MGM Team as the energy and sustainability manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that NV5 has assisted MGM MACAU and COTAI to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, Macao Green Hotel Award and China 3 Star Green Building Operation Label.

NV5 appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

1.0 PORTFOLIO PERFORMANCE SUMMARY

1.1 ELECTRICITY USE INTENSITY (EUI)

MACAU's electricity use intensity (EUI) in July 2023 had increased by **23.1%** compared to the same month in 2022 and increased by **7.9%** in 2023 compared to 2022. Meanwhile, COTAI's EUI in July 2023 had increased by **33.7%** compared to the same month in 2022 and increased by **12.2%** in 2023 compared to 2022.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	MACAU			COTAI		
	Year 2022 Actual EUI (kWh/m²)	Year 2023 Actual EUI (kWh/m²)	% Change	Year 2022 Actual EUI (kWh/m²)	Year 2023 Actual EUI (kWh/m²)	% Change
January	22.6	22.4	-0.8%	22.3	22.6	+1.6%
February	20.2	21.2	+5.2%	20.1	21.0	+4.5%
March	24.4	24.6	+1.0%	23.6	24.4	+3.3%
April	24.3	25.0	+3.0%	24.0	25.8	+7.3%
May	25.9	28.5	+9.9%	25.6	29.4	+14.7%
June	27.3	30.3	+11.1%	27.0	31.5	+17.0%
July	25.9	31.9	+23.1%	24.3	32.5	+33.7%
August						
September						
October						
November						
December						
Average	**24.4**	**26.3**	**+7.9%**	**23.8**	**26.7**	**+12.2%**

1.2 WATER USE INTENSITY (WUI)

MACAU's water use intensity (WUI) in July 2023 increased by **118.6%** compared to the same month in 2022 and increased by **29.2%** in 2023 compared to 2022. Meanwhile, COTAI's WUI in July 2023 increased by **149.5%** compared to the same month in 2022 and increased by **55.6%** for 2023 compared to 2022.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	MACAU			COTAI		
	Year 2022 Actual WUI (m^3/m^2)	Year 2023 Actual WUI (m^3/m^2)	% Change	Year 2022 Actual WUI (m^3/m^2)	Year 2023 Actual WUI (m^3/m^2)	% Change
January	0.18	0.19	+6.0%	0.12	0.14	+21.0%
February	0.20	0.21	+8.6%	0.12	0.16	+35.4%
March	0.19	0.21	+11.9%	0.12	0.17	+37.9%
April	0.21	0.24	+10.3%	0.14	0.19	+36.7%
May	0.23	0.28	+20.1%	0.15	0.21	+44.2%
June	0.19	0.29	+51.9%	0.12	0.23	+84.2%
July	0.15	0.32	+118.6%	0.10	0.25	+149.5%
August						
September						
October						
November						
December						
Average	**0.19**	**0.25**	**+29.2%**	**0.12**	**0.19**	**+55.6%**

2.0 MGM MACAU PERFORMANCE

2.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in July 2023 increased by 23.1% and increased by 21.7% respectively compared to the same month in 2022. To date, MGM MACAU has increased the actual electricity consumption by 7.9% and associated cost increased by 8.4% respectively.

Table 9. MACAU Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2022	2023	% Change Actual	2022	2023	% Change Actual	
January	4,784,280	4,745,400	-0.8%	5,519,350	5,668,020	+2.7%	-37.7%
February	4,266,240	4,488,360	+5.2%	4,942,590	5,365,140	+8.5%	+658.0%
March	5,152,320	5,205,240	+1.0%	5,938,160	6,207,870	+4.5%	-9.0%
April	5,131,680	5,284,800	+3.0%	6,183,420	6,236,780	+0.9%	-3.2%
May	5,478,840	6,019,800	+9.9%	6,595,030	7,106,470	+7.8%	+17.6%
June	5,777,040	6,418,080	+11.1%	7,567,480	8,307,460	+9.8%	+6.6%
July	5,480,880	6,748,440	+23.1%	7,172,780	8,730,530	+21.7%	-3.1%
August							
September							
October							
November							
December							
Total/Average	**36,071,280**	**38,910,120**	**+7.9%**	**43,918,810**	**47,622,270**	**+8.4%**	**+5.0%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on electricity consumption normalization.*

NV5 summarized the key items which could result in electricity consumption increase below:

- July 2022 was under COVID periods, thus majority of the utilities have increased significantly compared with the same month in 2023.
- Heat Pump Plant electricity consumption increased by 184.6% in July 2023 compared with the same month in 2022.
- Chiller Plant electricity consumption increased by 10.5% in July 2023 compared with the same month in 2022.
    - Chiller electricity consumption increased by 11.8% in July 2023 compared with the same month in 2022.
    - Cooling Tower electricity consumption increased by 14.0% in July 2023 compared with the same month in 2022.

Table 10 presents the comparison of electricity rates in the years 2022 and 2023. The electricity rate in July 2023 decreased by 1.1% compared to the same month in 2022.

Table 10. MACAU - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2022 – Electricity Rate (MOP/kWh)	Year 2023 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.15	1.19	+3.5%
February	1.16	1.20	+3.2%
March	1.15	1.19	+3.5%
April	1.20	1.18	-2.1%
May	1.20	1.18	-1.9%
June	1.31	1.29	-1.2%
July	1.31	1.29	-1.1%
August			
September			
October			
November			
December			
Total/Average	**1.21**	**1.22**	**+0.5%**

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in July 2023 decreased by 0.6% compared to 2019 normalized consumption in the same month.

Table 11. MACAU - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2023 Actual Electricity Consumption (kWh)	% Change – Normalized
January	4,715,548	4,745,400	+0.6%
February	4,893,297	4,488,360	-8.3%
March	5,275,299	5,205,240	-1.3%
April	5,597,956	5,284,800	-5.6%
May	6,166,350	6,019,800	-2.4%
June	6,591,853	6,418,080	-2.6%
July	6,788,616	6,748,440	-0.6%
August			
September			
October			
November			
December			
Total/Average	**40,028,920**	**38,910,120**	**-2.8%**

2.2 LPG CONSUMPTION

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, NV5 focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption in July 2023 increased by **49.9%** and cost increased **30.4%** respectively compared to the same month in 2022. To date, MGM MACAU has decreased the actual LPG consumption by **11.4%** and associated costs decreased by **19.1%.**

Table 12. MACAU – Summary of Actual Monthly LPG Consumption and Cost

Monthly Period	Actual LPG Consumption (kg)			Actual LPG Cost (MOP)			% Change – Min OAT
	2022	2023	% Change Actual	2022	2023	% Change Actual	
January	121,221	123,108	+1.6%	1,327,490	1,084,579	-18.3%	-13.7%
February	146,623	78,483	-46.5%	1,450,103	768,347	-47.0%	+105.6%
March	91,040	77,597	-14.8%	900,382	759,672	-15.6%	+5.7%
April	85,743	77,139	-10.0%	858,756	755,191	-12.1%	+42.3%
May	82,704	80,383	-2.8%	992,447	839,195	-15.4%	+15.4%
June	77,044	71,232	-7.5%	924,531	743,662	-19.6%	+2.8%
July	44,991	67,441	+49.9%	539,887	704,081	+30.4%	+2.7%
August							
September							
October							
November							
December							
Total/Average	**649,365**	**575,382**	**-11.4%**	**6,993,595**	**5,654,727**	**-19.1%**	**+14.5%**

Important note: *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

NV5 summarized the key items which could result in LPG consumption increase below:

- July 2022 was under COVID periods, thus majority of the utilities have increased significantly compared with the same month in 2023.
- Boiler LPG consumption increased by 27.0% in July 2023 compared with the same month in 2022.
- Kitchen LPG consumption increased by 304.7% in July 2023 compared with the same month in 2022.

Table 13 presents the comparison of LPG rates in the years 2022 and 2023. The LPG rate in July 2023 decreased by 13.0% compared with the same month in 2022.

Table 13. MACAU - Summary of Actual Monthly LPG Rate

Monthly Period	Year 2022 – LPG Rate (MOP/kg)	Year 2023 – LPG Rate (MOP/kg)	% Change – LPG Rate
January	10.95	8.81	-19.6%
February	9.89	9.79	-1.0%
March	9.89	9.79	-1.0%
April	10.02	9.79	-2.3%
May	12.00	10.44	-13.0%
June	12.00	10.44	-13.0%
July	12.00	10.44	-13.0%
August			
September			
October			
November			
December			
Total/Average	**10.96**	**9.93**	**-9.4%**

Table 14 summarizes the monthly normalized LPG consumption. The actual LPG consumption in July 2023 decreased by 17.0% compared to 2019 normalized consumption in the same month.

Table 14. MACAU - Summary of Normalized Monthly LPG Consumption

Monthly Period	2019 Normalized LPG Consumption (kg)	2023 Actual LPG Consumption (kg)	% Change – Normalized
January	103,641	123,108	+18.8%
February	84,783	78,483	-7.4%
March	77,414	77,597	+0.2%
April	76,016	77,139	+1.5%
May	80,862	80,383	-0.6%
June	82,361	71,232	-13.5%
July	81,206	67,441	-17.0%
August			
September			
October			
November			
December			
Total/Average	**586,283**	**575,382**	**-1.9%**

2.3 ISO50001 ENERGY CONSUMPTION RECORD

The MACAU ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to July 2023 is **5.4%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to the 2019 baseline.

Table 15. MACAU ISO 50001 Normalized Energy Performance Targets for 2025

MGM Property	Baseline	Energy Savings	2021	2022	Normalized Savings Achieved Up to July 2023	2024	2025
MACAU	2019	Target	-4%	-8%	-10%	N/A	**-13%**
		Actual	**-5.9%**	**-10.8%**	**-5.4%**	-	**-**

Table 16. MACAU ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2023 Actual Energy Consumption (GJ)	% Change – Normalized
January	23,023	22,660	-1.6%
February	23,041	19,713	-14.4%
March	22,946	22,254	-3.0%
April	25,252	22,520	-10.8%
May	25,652	25,313	-1.3%
June	27,685	26,332	-4.9%
July	28,115	27,349	-2.7%
August			
September			
October			
November			
December			
Total/Average	**175,714**	**166,141**	**-5.4%**

Important note: *Factors used for unit conversion: Electricity (0.0036) and LPG (0.0453)*

2.4 WATER CONSUMPTION

Based on the existing water metering system, the overall site water consumption can be broken down into five (5) categories: Kitchen and Restaurants; Hotel Tower; Fire Services; Chiller Plant & miscellaneous.

According to Table 17, the actual water consumption and cost in July 2023 has increased by 118.6% and 118.4% compared to the same month in 2022 respectively. To date, MGM MACAU has increased both the actual water consumption and associated cost by 29.2% and 29.1%.

Table 17. MACAU - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2022	2023	% Change Actual	2022	2023	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	37,078	39,305	+6.0%	287,500	305,020	+6.1%	-37.7%	+52.7%
February	41,524	45,112	+8.6%	322,270	350,080	+8.6%	+658.0%	+16.6%
March	39,962	44,725	+11.9%	310,160	347,050	+11.9%	-9.0%	+71.8%
April	45,295	49,960	+10.3%	351,540	387,650	+10.3%	-3.2%	+87.5%
May	49,042	58,878	+20.1%	380,590	456,800	+20.0%	+17.6%	+77.2%
June	40,383	61,331	+51.9%	313,390	475,780	+51.8%	+6.6%	+195.1%
July	31,179	68,165	+118.6%	242,100	528,630	+118.4%	-3.1%	+2472.4%
August								
September								
October								
November								
December								
Total/Average	**284,463**	**367,476**	**+29.2%**	**2,207,550**	**2,851,010**	**+29.1%**	**+5.0%**	**+107.7%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on water consumption normalization.*

NV5 summarized the key items which could result in water consumption increase below:

- July 2022 was under COVID periods, thus majority of the utilities have increased significantly compared with the same month in 2023.
- Hotel Tower water consumption for July 2023 has increased by 320.6% compared to the same month in 2022.
- Kitchen water consumption for July 2023 has increased by 105.0% compared to the same month in 2022.
- CT Make Up water consumption for July 2023 has increased by 21.7% compared to the same month in 2022.

Table 18 presents the comparison of water rates in the years 2022 and 2023. The water rate in July 2023 has decreased by 0.12% compared to the same month in 2022. The average utility rate of water remains unchanged compared to 2022.

Table 18. MACAU - Summary of Actual Monthly Water Rates

Monthly Period	Year 2022 – Water Rate (MOP/m³)	Year 2023 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.754	7.760	+0.08%
February	7.761	7.760	-0.01%
March	7.761	7.760	-0.02%
April	7.761	7.759	-0.02%
May	7.760	7.758	-0.03%
June	7.760	7.758	-0.04%
July	7.765	7.755	-0.12%
August			
September			
October			
November			
December			
Total/Average	**7.760**	**7.759**	**-0.02%**

Table 19 summarized the monthly normalized water consumption. The actual water consumption in July 2023 increased by 0.8% compared to 2019 normalized consumption in the same month.

Table 19. MACAU - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2023 Actual Water Consumption (m³)	% Change – Normalized
January	43,437	39,305	-9.5%
February	42,590	45,112	+5.9%
March	47,450	44,725	-5.7%
April	50,917	49,960	-1.9%
May	55,459	58,878	+6.2%
June	59,235	61,331	+3.5%
July	67,617	68,165	+0.8%
August			
September			
October			
November			
December			
Total/Average	**366,705**	**367,476**	**+0.2%**

2.5 MACAU YEAR 2023 PROJECTS

NV5 will continue to perform analysis and coordinate with MGM to implement the following projects in 2023 and reduce the site energy and water consumption which supports NV5 annual normalized energy savings target.

Table 20. MGM MACAU Year 2023 Projects

Item	Initiatives	Status
Electricity		
1	Secondary Chilled Water Pump Retrofit (Phase 1)	Postpone to 2024
2	Cooling Tower Fan Retrofit (Phase 1)	On-going
3	PICCV Implementation 2023 (Grande Praça 7 Units)	Completed
4	Neon Light LED Replacement (Phase 3)	Postpone to 2024
5	Hotel Tower Staircase Lighting Delamping	Completed
6	Rooftop Façade LED Replacement	On-going
7	Lion Backdrop Lighting Upgrade	On-going
LPG		
8	Natural Gas Changeover Project	On-going

3.0 MGM COTAI PERFORMANCE

3.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in July 2023 increased by 33.7% and 31.6% respectively compared to the same month in 2022. To date, MGM COTAI has increased the actual electricity consumption by 12.2% and associated cost increased by 12.4%.

Table 21. COTAI Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2022	2023	% Change Actual	2022	2023	% Change Actual	
January	8,336,400	8,471,280	+1.6%	10,013,460	10,516,100	+5.0%	-37.7%
February	7,528,560	7,869,600	+4.5%	9,066,060	9,794,150	+8.0%	+658.0%
March	8,833,200	9,128,520	+3.3%	10,632,040	11,334,370	+6.6%	-9.0%
April	8,985,360	9,638,280	+7.3%	11,262,370	11,855,350	+5.3%	-3.2%
May	9,585,000	10,993,680	+14.7%	11,983,070	13,528,490	+12.9%	+17.6%
June	10,087,080	11,802,600	+17.0%	12,615,080	14,491,170	+14.9%	+6.6%
July	9,082,680	12,146,160	+33.7%	11,346,140	14,926,180	+31.6%	-3.1%
August							
September							
October							
November							
December							
Total/Average	**62,438,280**	**70,050,120**	**+12.2%**	**76,918,220**	**86,445,810**	**+12.4%**	**+5.0%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on electricity consumption normalization*

NV5 summarized the key items which could result in electricity consumption increase below:

- July 2022 was under COVID periods, thus majority of the utilities have increased significantly compared with the same month in 2023.
- Chiller Plant electricity consumption for July 2023 has increased by 31.5% compared to the same month in 2022.
- Kitchen & Restaurant electricity consumption for July 2023 has increased by 110.2% compared to the same month in 2022.
- Hotel Towers electricity consumption for July 2023 has increased by 57.9% compared to the same month in 2022.
- Casino electricity consumption for July 2023 has increased by 36.3% compared to the same month in 2022.

Table 22 presents the monthly e-bus electricity consumption in MGM COTAI. The e-bus was not operated during July 2022 COVID periods, there is an absolute 148,375 kWh e-bus electricity increase in July 2023.

Table 22. COTAI - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2022	2023	% Change – Actual
January	42,287	51,261	+21.2%
February	47,905	53,640	+12.0%
March	38,793	66,865	+72.4%
April	38,537	72,669	+88.6%
May	51,328	132,741	+158.6%
June	36,420	140,991	+287.1%
July	0	148,375	-
August			
September			
October			
November			
December			
Total/Average	**255,269**	**666,542**	**+161.1%**

Table 23 presents the comparison of electricity rates in the years 2022 and 2023. The electricity rate in July 2023 decreased by 1.6% compared to the same month in 2022.

Table 23. COTAI - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2022 – Electricity Rate (MOP/kWh)	Year 2023 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.20	1.24	+3.3%
February	1.20	1.24	+3.3%
March	1.20	1.24	+3.2%
April	1.25	1.23	-1.9%
May	1.25	1.23	-1.6%
June	1.25	1.23	-1.8%
July	1.25	1.23	-1.6%
August			
September			
October			
November			
December			
Total/Average	**1.230**	**1.235**	**+0.4%**

Table 24 summarizes the monthly normalized electricity consumption. The actual electricity consumption in July 2023 increased by **4.1%** compared to 2019 normalized consumption in the same month.

Table 24. COTAI - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2023 Actual Electricity Consumption (kWh)	% Change – Normalized
January	8,521,687	8,471,280	-0.6%
February	8,791,529	7,869,600	-10.5%
March	9,371,451	9,128,520	-2.6%
April	9,861,279	9,638,280	-2.3%
May	10,724,164	10,993,680	+2.5%
June	11,370,125	11,802,600	+3.8%
July	11,668,833	12,146,160	+4.1%
August			
September			
October			
November			
December			
Total/Average	**70,309,068**	**70,050,120**	**-0.4%**

3.2 NATURAL GAS CONSUMPTION

The actual natural gas consumption and cost in July 2023 increased by 2184.3% and 2391.9% respectively compared to the same month in 2022, respectively. To date, MGM COTAI has increased actual natural gas consumption by 43.0% and associated costs by 53.0%.

Table 25. COTAI - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual NG Consumption (m^3)			Actual NG Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2022	2023	% Change Actual	2022	2023	% Change Actual		
January	59,378	64,043	+7.9%	376,240	405,800	+7.9%	-13.7%	+67.3%
February	76,089	58,053	-23.7%	482,130	378,840	-21.4%	+105.6%	+29.8%
March	52,146	63,461	+21.7%	330,410	438,660	+32.8%	+5.7%	+135.0%
April	43,900	62,658	+42.7%	278,170	433,120	+55.7%	+42.3%	+184.5%
May	45,632	66,163	+45.0%	289,140	457,350	+58.2%	+15.4%	+197.2%
June	33,949	65,456	+92.8%	215,120	452,450	+110.3%	+2.8%	+373.6%
July	3,041	69,467	+2184.3%	19,270	480,190	+2391.9%	+2.7%	+3870.4%
August								
September								
October								
November								
December								
Total/Average	**314,135**	**449,301**	**+43.0%**	**1,990,480**	**3,046,410**	**+53.0%**	**+14.5%**	**+174.0%**

Important note: *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization*

NV5 summarized the key items which could result in natural gas consumption increase below:

- July 2022 was under COVID periods, thus majority of the utilities have increased significantly compared with the same month in 2023.
- Kitchen NG consumption for July 2023 has increased by 1830.2% compared to the same month in 2022.
- Boiler NG consumption for July 2023 has increased by 340.0% compared to the same month in 2022.

Table 26 presents the comparison of natural gas rates in the years 2022 and 2023. The natural gas rate in July 2023 increased by 9.1% compared to the same month in 2022.

Table 26. COTAI - Summary of Actual Monthly Natural Gas Rate

Monthly Period	Year 2022 – Natural Gas Rate (MOP/m³)	Year 2023 – Natural Gas Rate (MOP/m³)	% Change – NG Rate
January	6.34	6.34	-
February	6.34	6.53	+3.0%
March	6.34	6.91	+9.1%
April	6.34	6.91	+9.1%
May	6.34	6.91	+9.1%
June	6.34	6.91	+9.1%
July	6.34	6.91	+9.1%
August			
September			
October			
November			
December			
Total/Average	**6.34**	**6.77**	**+6.9%**

Table 27 summarizes the monthly normalized natural gas consumption. The actual natural gas consumption in July 2023 decreased by 32.3% compared to 2019 normalized consumption in the same month.

Table 27. COTAI - Summary of Normalized Monthly Natural Gas Consumption

Monthly Period	2019 Normalized Natural Gas Consumption (m³)	2023 Actual Natural Gas Consumption (m³)	% Change – Normalized
January	128,174	64,043	-50.0%
February	120,801	58,053	-51.9%
March	117,977	63,461	-46.2%
April	113,428	62,658	-44.8%
May	112,330	66,163	-41.1%
June	104,173	65,456	-37.2%
July	102,604	69,467	-32.3%
August			
September			
October			
November			
December			
Total/Average	**799,486**	**449,301**	**-43.8%**

3.3 ISO50001 ENERGY CONSUMPTION RECORD

COTAI ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to July 2023 is **4.9%** as shown in Table 28 and Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

Table 28. COTAI ISO 50001 Normalized Energy Performance Targets for 2025

MGM Property	Baseline	Energy Savings	2021	2022	Normalized Savings Achieved Up to July 2023	2024	2025
COTAI	2019	Target	-4%	-8%	-10%	N/A	**-13%**
		Actual	**-6.4%**	**-16.9%**	**-4.9%**	-	**-**

Table 29. COTAI ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2023 Actual Energy Consumption (GJ)	% Change – Normalized
January	35,567	32,719	-8.0%
February	36,708	30,345	-17.3%
March	37,322	35,065	-6.0%
April	40,715	36,872	-9.4%
May	41,466	41,873	+1.0%
June	44,419	44,761	+0.8%
July	45,330	46,137	+1.8%
August			
September			
October			
November			
December			
Total/Average	**281,527**	**267,771**	**-4.9%**

***Important note**: Factors used for unit conversion: Electricity (0.0036) and Natural Gas (0.0347)*

3.4 WATER CONSUMPTION

According to Table 30, the actual water consumption and associated costs in July 2023 increased by **149.5%** and **149.0%** respectively compared to the same month in 2022. To date, MGM COTAI has increased actual water consumption by **55.6%**, and associated costs by **55.5%**.

Table 30. COTAI - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2022	2023	% Change Actual	2022	2023	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	44,658	54,018	+21.0%	347,240	419,790	+20.9%	-37.7%	+67.3%
February	45,298	61,317	+35.4%	352,330	476,420	+35.2%	+658.0%	+29.8%
March	44,914	61,936	+37.9%	349,260	481,140	+37.8%	-9.0%	+135.0%
April	51,640	70,617	+36.7%	401,500	548,440	+36.6%	-3.2%	+184.5%
May	55,628	80,211	+44.2%	432,480	622,920	+44.0%	+17.6%	+197.2%
June	45,805	84,354	+84.2%	356,090	654,900	+83.9%	+6.6%	+373.6%
July	38,031	94,905	+149.5%	295,890	736,800	+149.0%	-3.1%	+3870.4%
August								
September								
October								
November								
December								
Total/Average	**325,974**	**507,358**	**+55.6%**	**2,534,790**	**3,940,410**	**+55.5%**	**+5.0%**	**+174.0%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization*

NV5 summarized the key items which could result in water consumption increased below:

- July 2022 was under COVID periods, thus majority of the utilities have increased significantly compared with the same month in 2023.
- Hotel Towers and Mansion water consumption for July 2023 has increased by 743.6% and 80.7% compared with the same month last year.
- CT Make Up water consumption for July 2023 has increased by 37.6% compared with the same month last year.
- Kitchen water consumption for July 2023 has increased by 69.7% compared with the same month last year.

Table 31 presents the comparison of water rates in the years 2022 and 2023. The water rate in July 2023 decreased by 0.2% compared to the same month in 2022. To date, the average utility rate of water has decreased by 0.1% compared to 2022.

Table 31. COTAI - Summary of Actual Monthly Water Rates

Monthly Period	Year 2022 – Water Rate (MOP/m³)	Year 2023 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.776	7.771	-0.05%
February	7.778	7.770	-0.1%
March	7.776	7.768	-0.1%
April	7.775	7.766	-0.1%
May	7.774	7.764	-0.1%
June	7.774	7.764	-0.1%
July	7.780	7.764	-0.2%
August			
September			
October			
November			
December			
Total/Average	**7.776**	**7.767**	**-0.1%**

Table 32 summarized the monthly normalized water consumption. The actual water consumption in July 2023 increased by 10.4% compared to 2019 normalized consumption in the same month.

Table 32. COTAI - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2023 Actual Water Consumption (m³)	% Change – Normalized
January	57,914	54,018	-6.7%
February	56,894	61,317	+7.8%
March	62,913	61,936	-1.6%
April	69,294	70,617	+1.9%
May	75,963	80,211	+5.6%
June	79,803	84,354	+5.7%
July	85,984	94,905	+10.4%
August			
September			
October			
November			
December			
Total/Average	**488,764**	**507,358**	**+3.8%**

3.5 COTAI YEAR 2023 PROJECTS

NV5 will continue to perform analysis and coordinate with MGM to implement the following projects in 2023 and reduce the site energy and water consumption which supports NV5 annual normalized energy savings target.

Table 33. MGM COTAI Year 2023 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Plant Controls Optimization (CHWP DP Setpoint Reset, CHWST Setpoint Reset)	On-going
2	PICCV Implementation 2023 (12 Units)	Postpone to 2024
3	KEF Steps Speed Control Optimization (Vista, Ball Room, MDM, Hao Guo, Five Foot Road, Chún, Coast)	On-going
4	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)	On-going
5	Centralized RO Pump Set Replacement (10 units)	Completed
6	Booster Pump Set Replacement (2 units)	Completed
Water		
7	Extension of B2 Carpark CDP Water Recycling (Phase 5 to 7)	On-going
8	Spectacle Façade Rainwater Harvesting for CT Make Up	TBC

APPENDIX A

Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target for MGM MACAU and MGM COTAI. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2023.

MGM MACAU

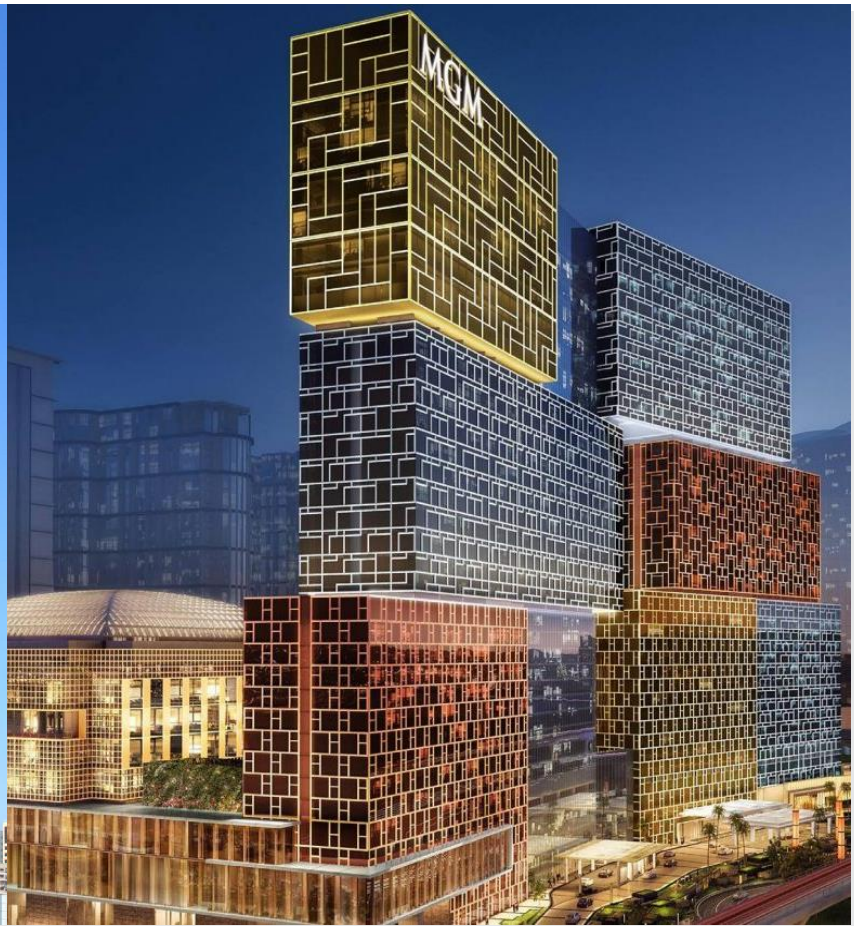
Utilities	2023	
Electricity (kWh)	CDD (Base T=17°C)	
LPG (kg)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM COTAI

Utilities	2023	
Electricity (kWh)	CDD (Base T=17°C)	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM MACAU & COTAI
Resource Efficiency Management Report
860720-0006650.00
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